



密集追踪数据的亚组分析： 组迭代多模型估计

Subgrouping in Intensive Longitudinal Data Analysis: Group Iterative Multiple Model Estimation (GIMME)

唐莹莹 TANG Yingying

Human Development and Family Sciences Department, University of
Texas at Austin

5/8/2024

Some simple but complicated questions...

1. What are your typical behaviors when you use your smart phones?

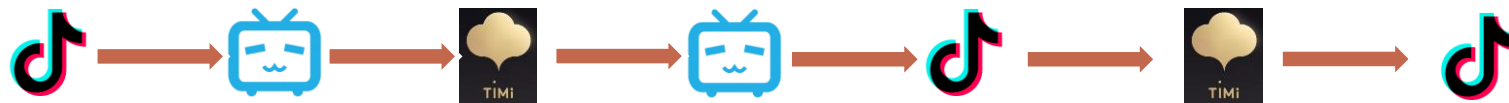
- Some people reply to Wechat messages, lock their phones and go back to studying



- Some people open 小红书 to search for answers but never leave For You page



- Some people switch between Tiktok, Bilibili, and TiMi

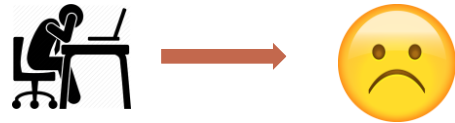


- Are people who show the first behavior pattern also the people who show the second behavior pattern?

Some simple but complicated questions...

2. In daily life, do people all react the same from stress events?

- Some people feel bad the day they perceive high stress, but other people don't



- Some people's negative feelings carry over to the next day, but other people don't



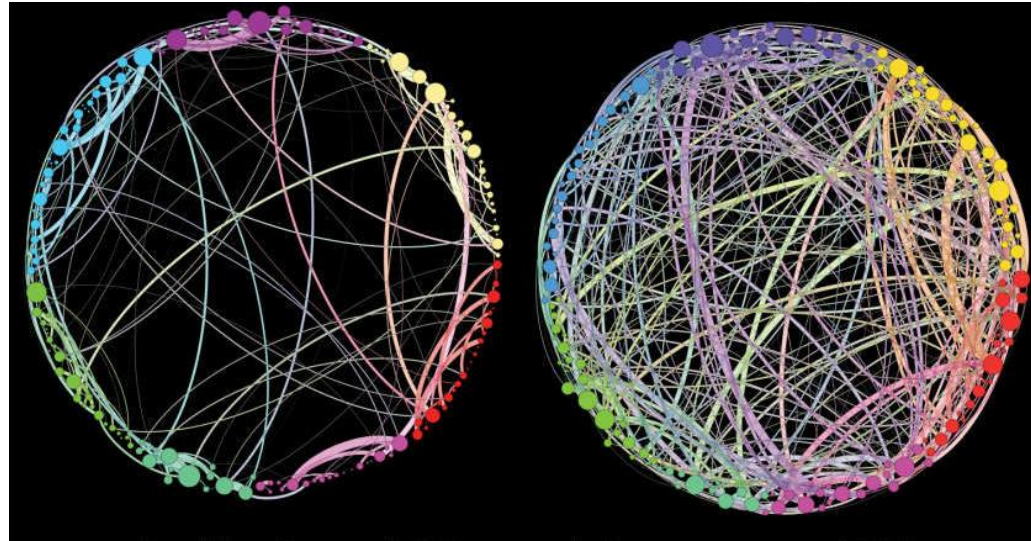
- Some people experience more stressful events because they feel bad, but other people don't



- Are people who show the first behavior pattern also the people who show the second behavior pattern?

Some simple but complicated questions...

3. Do people show the same brain connectivity when watch a sad movie?



How to use statistical methods to answer these questions?

Data Driven Method

Group Iterative Multiple Model Estimation (GIMME)

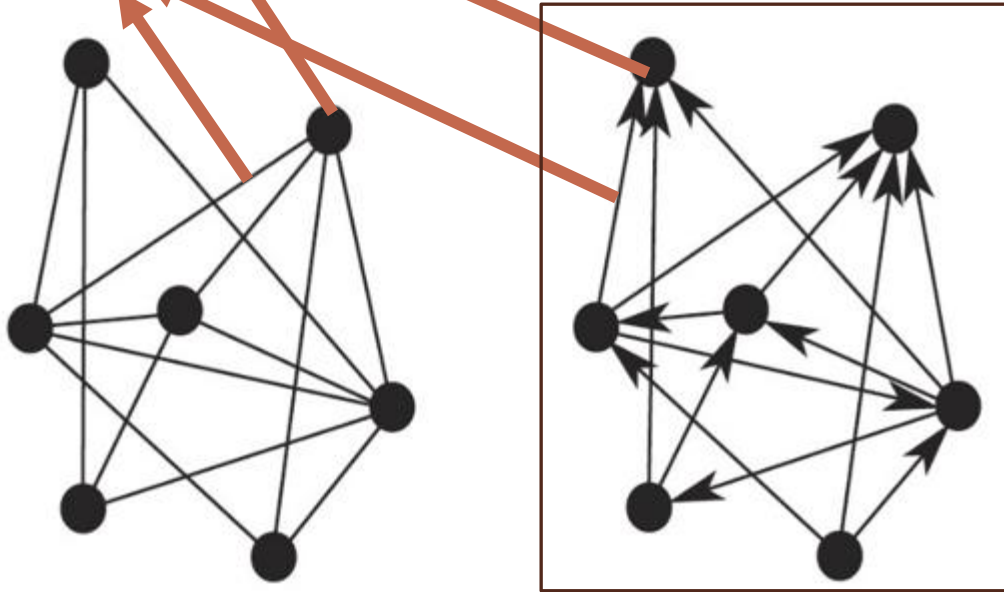
- An algorithm for finding models that describe dynamic processes
- statistical models (i.e., networks) that explain patterns of temporal covariation in a system

Subgrouping Group Iterative Multiple Model Estimation (S-GIMME)

- general patterns of relations that exist for the entire sample
- patterns that are shared by only a subset of individuals
- patterns that are unique to a given individual

Let's start from network...

Nodes Components
Edge Links between nodes



General representation

observable variables
e.g., partial correlations



neural networks



social networks

A little math behind GIMME

Unified structural equation models (uSEM)

- a type of structural vector autoregressive (VAR) model: a class of models that contain both contemporaneous and lagged parameters
- identify both lagged and contemporaneous relations in the same step, producing results independent of variable input order

Diagram illustrating the Equation for GIMME:

The equation is:

$$\eta_i(t) = (A_i + A_i^g) \eta_i(t) + (\Phi_{1,i} + \Phi_{1,i}^g) \eta_i(t-1) + \zeta_i(t).$$

Annotations:

- $\eta_i(t)$: observed time series
- i : i^{th} individual
- $A_i + A_i^g$: $p \times p$ matrix of contemporaneous edge (diagonal = 0)
- $\Phi_{1,i} + \Phi_{1,i}^g$: matrix of first order lagged edges (diagonal = autoregressive effects)
- g : group
- $\zeta_i(t)$: p -variate vector of errors assumed to have means of zero with a diagonal covariance matrix and no sequential dependencies, such that all temporal information is explained within A and Φ_1

Equation for GIMME (from Beltz & Gates, 2017)

Diagram illustrating the Equation for S-GIMME:

The equation is:

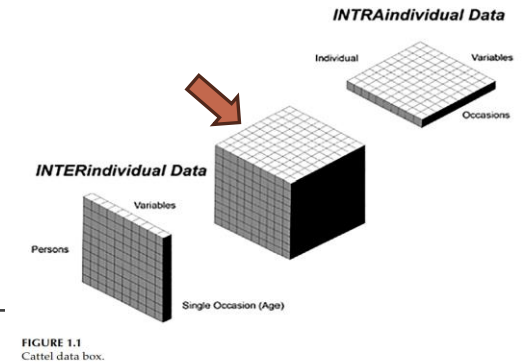
$$\eta_{i,t} = (A_i + A_{i,k}^s + A_i^g) \eta_{i,t} + (\phi_i + \phi_{i,k}^s + \phi_i^g) \eta_{i,t-1} + \zeta_{i,t}$$

Annotations:

- $\eta_{i,t}$: observed time series
- s : subgroup

Equation for S-GIMME (from Lane et al., 2019)

Requirements for the data



multivariate time series data

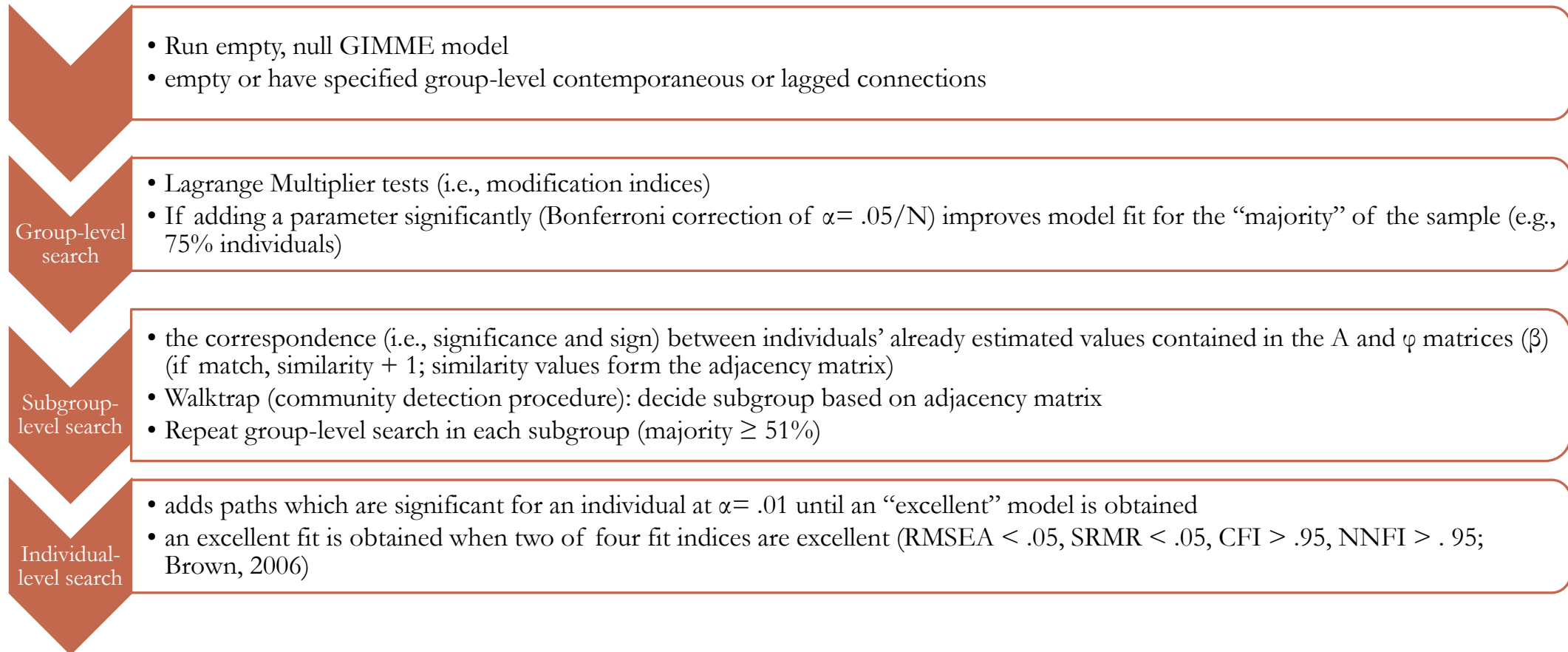
Assumptions of GIMME: Stationary (must) and Ergodicity (S-GIMME doesn't require all individuals have the same process)

- Weak stationary: mean, variance, and covariance, that are constant over time and only depend on the lag
- If data has a trend, remove the trend and use residuals (Beltz & Gates, 2017)
- General requirement for SEM model

Rules of thumb

1. variables 3-25 (5-15 recommended)
2. recommend at least 100 time points ($T = 60$ and 5 or 10 variables may be adequate)
3. rows with missing data < rows with data (else may not converge; GIMME package use GIMME utilizes FIML to deal with missing data; missing data can be imputed)
4. GIMME at least **10** individuals; S-GIMME at least **25** individuals (recommended)

Process of S-GIMME



Process of S-GIMME

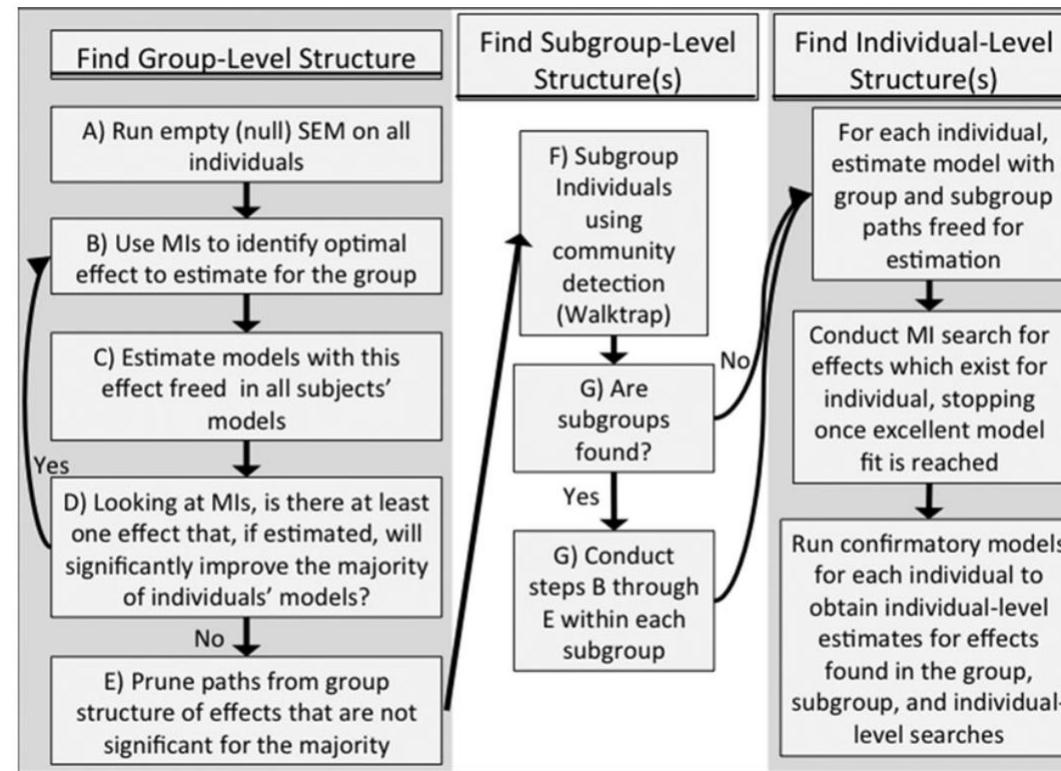
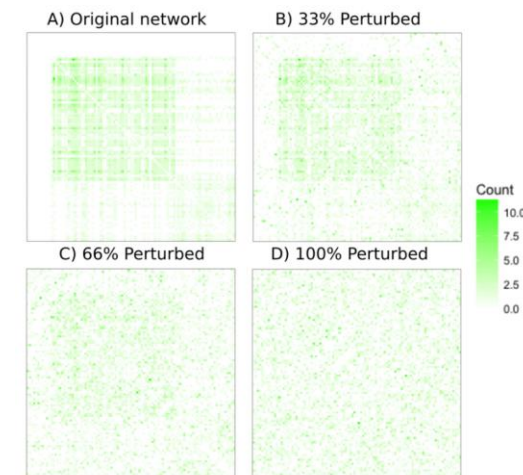


Figure 3. Schema for subgrouping within GIMME (S-GIMME).

Evaluation of S-GIMME

Are the results accurate/robust?

- **Subgrouping:** A **modularity value** (Q), the degree to which similarity with others within a subgroup is high relative to the degree of similarity between subgroups
 - Q = the number of edges falling within groups - the expected number in an equivalent network with edges placed at random
 - **not appropriate** when evaluating **absolute** quality of subgroup partitions (e.g., changed dataset)
 - $Q > 0$, proceed with path evaluation; $Q < 0$, not robust
- **Paths:** **perturbing** based on the adjacency matrix
 - Step 1: Randomly perturb a specific proportion of edges (α)
to rewire the original matrix
 - Step 2: Obtain a cluster solution for the rewired matrix
 - Step 3: Compare two solutions using VI (Variation of Information) or ARI (Hubert-Arabie Adjusted Rand Index)
 - ROT: $\geq 20\%$ edges to be perturbed before 20% of individuals in different clusters



Applications

1. fMRI studies

- First study: college students complete go/no-go task with alcoholic or non-alcoholic beverages

2. Ecological Momentary Assessment (EMA) studies

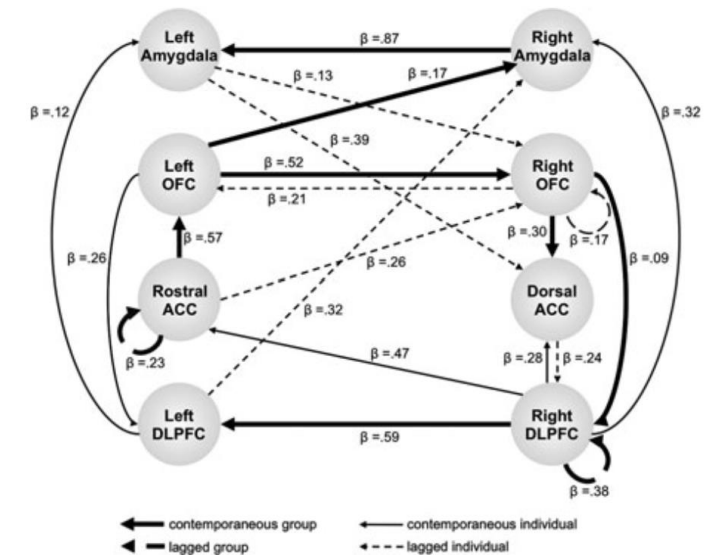
- How emotion, behavior, and cognition unfold in the naturalistic settings of daily life

3. passively collected data (e.g., via smart phone)

- Example in the beginning

4. fNIRS studies (hyperscanning)

- The temporal link of brain activity and behaviors both within and across individuals



Kullar et al., 2024 *Emotion*



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ISSN: 1528-3542

Emotion

2024, Vol. 24, No. 1, 116–129
<https://doi.org/10.1037/emo0001245>

Patterns of Emotion-Network Dynamics Are Orthogonal to Mood Disorder Status: An Experience Sampling Investigation

Monica Kullar¹, Scott Carter², Caitlin Hitchcock^{1, 3}, Steve Whittaker⁴, Aidan G. C. Wright⁵, and Tim Dalgleish^{1, 6}

¹ Medical Research Council Cognition and Brain Sciences Unit, University of Cambridge

² Targhetta Design, LLC, San Jose, California, United States

³ Melbourne School of Psychological Sciences, University of Melbourne

⁴ Department of Computational Media, University of California

⁵ Department of Psychology, University of Pittsburgh

⁶ Cambridgeshire and Peterborough NHS Foundation Trust, Cambridge, Cambridgeshire, United Kingdom

Background

People with mood disorders (e.g., depression, bipolar disorder) are compared with healthy people in previous research.

In addition to *between-group* differences, *within-person* dynamics in behavior, affect, and cognition and their interaction with the environment may vary across individuals. → *interventions that disrupt dysfunctional dynamics*

Compatible with transdiagnostic approach (e.g., RDoC and HiTOP)

Current Study

Unfolding emotional experiences in 14 days

A diverse sample: (a) major depression currently in episode, (b) major depression in remission, (c) bipolar disorder, or (d) no history of mood disorder

1. Whether subgroups based on data-driven **pattern in emotional dynamics** align with **traditional diagnostic** grouping → confirmatory GIMME and data-driven GIMME
2. Potential differences on demographic (e.g., age, gender), clinical variables (e.g., symptom frequency, intensity), and emotion metrics (not discussed today; emotion intensity, instability, inertia, and granularity)

Measurement and Procedure

Emotion self-report: enthusiastic, happy, pleased, relaxed, nervous, sad, angry, irritated, and stressed

Combined emotions due to high correlation

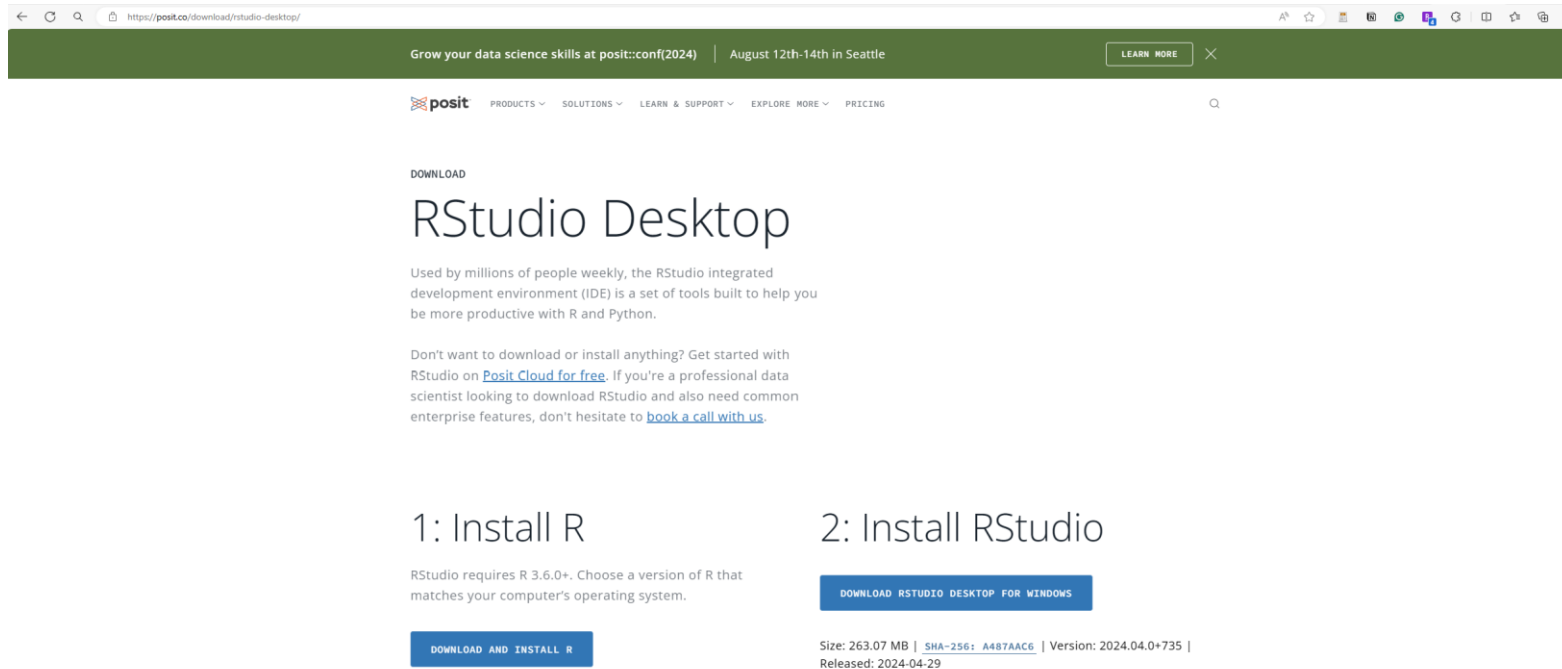
- Happy + pleased → Happy
- Angry + irritated → Angry

SMS: starting from 10:00 am, every 2 hr, 5 times a day, 14 days

Random time within a 30-min window

Replicate (Rerun) the analysis

Install R and Rstudio: Go to Rstudio website (<https://posit.co/downloads/>)



The screenshot shows the RStudio Desktop download page on the Posit website. The browser address bar shows <https://posit.co/download/rstudio-desktop/>. The page has a green header with a navigation bar containing links for PRODUCTS, SOLUTIONS, LEARN & SUPPORT, EXPLORE MORE, and PRICING. A green banner at the top promotes 'posit::conf(2024)' on August 12th-14th in Seattle, with a 'LEARN MORE' button. The main content area is titled 'DOWNLOAD RStudio Desktop' and describes the IDE. It includes a section for '1: Install R' with a 'DOWNLOAD AND INSTALL R' button, and a section for '2: Install RStudio' with a 'DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS' button. Technical details for the Windows download are provided at the bottom right.

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RStudio Desktop

Used by millions of people weekly, the RStudio integrated development environment (IDE) is a set of tools built to help you be more productive with R and Python.

Don't want to download or install anything? Get started with RStudio on [Posit Cloud for free](#). If you're a professional data scientist looking to download RStudio and also need common enterprise features, don't hesitate to [book a call with us](#).

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

[DOWNLOAD AND INSTALL R](#)

2: Install RStudio

[DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS](#)

Size: 263.07 MB | [SHA-256: A487AAC6](#) | Version: 2024.04.0+735 | Released: 2024-04-29

Roadmap for analysis

1. Basic R and Data preprocessing (introduce Rstudio, import data, data wrangling, descriptive analysis, correlation)
2. Confirmatory GIMME (GIMME package in R), evaluation of robustness
3. Exploratory GIMME (GIMME package in R) , evaluation of robustness
4. The differences in demographic and clinical metrics (skip)

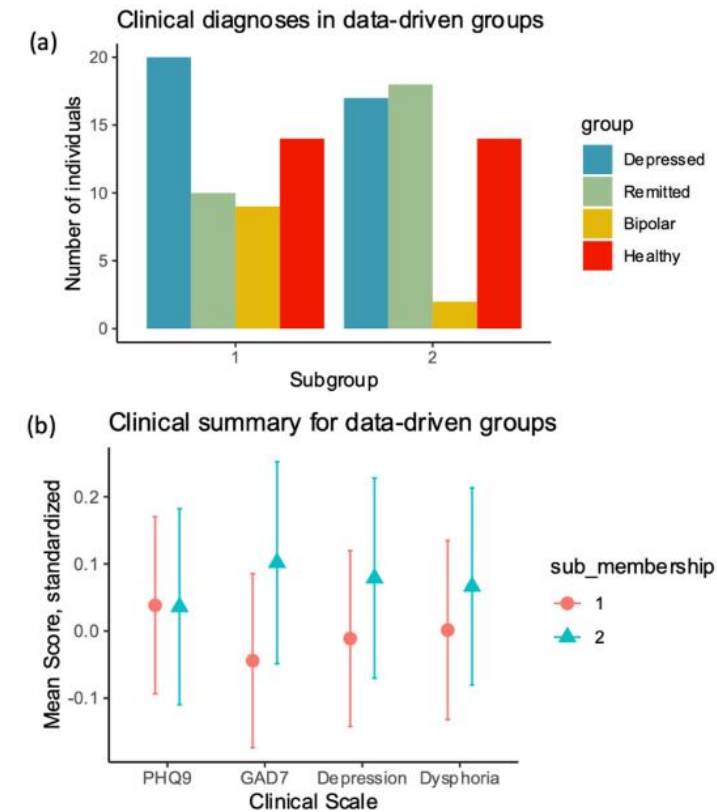
Results

	Subgroup 1	Subgroup 2	Statistic
n	53	51	
Demographics			
Age, years: <i>M (SD)</i>	41.96 (11.78)	38.62 (15.24)	$F=1.57, p=.21$
Female	37 (69.81%)	39 (76.47 %)	$p=.51^*$
Male	16 (30.19%)	12 (23.53 %)	
Ethnicity			
			$p=.27^*$
White (British)	45 (84.91%)	48 (94.12%)	
White (other)	4 (7.55%)	1 (1.96%)	
Asian	3 (5.66%)	1 (1.96%)	
Black	-	1 (1.96%)	
Mixed White/Asian	1 (1.89%)	-	
Income (£)			
			$p=.99^*$
<10,000	11 (20.75%)	12 (23.53%)	
10,000-29,999	24 (45.28%)	24 (47.06%)	
30,000-49,999	9 (16.98%)	7 (13.73%)	
50,000-69,999	3 (5.66%)	2 (3.92%)	
Preferred not to say	6 (11.32%)	6 (11.76%)	
Highest Education Level			
			$p=.91^*$
GCSE	4 (87.55%)	5 (9.80%)	
A-levels	6 (11.32%)	9 (17.65%)	
HND/BTEC/NVQ levels	7 (13.02%)	7 (13.73%)	
Bachelor's degree	21 (39.62%)	17 (33.33%)	
Master's degree	10 (18.87%)	10 (19.61%)	
Doctorate degree	5 (9.43%)	3 (5.88%)	

No differences in demographics

Figure 2

Clinical Characteristics for the Two Data-Driven Subgroups



No differences in clinical characteristics

Limitation of (S-)GIMME

1. Data-driven

- the possibility of overfitting the data

2. High requirements for data, especially when using EMA

- Biased sample
- Cost
- Variation
- Unequal time interval (Kullar used missingness to replace night-time, any problems?)
- Lags > 1 ?



Take Home Messages

1. (S-)GIMME, a data-driven method to find models that describe dynamic processes
2. S-GIMME generates group- , subgroup- , and individual-level results
3. GIMME requires many measurement points
4. *gimme* and *perturbR* packages in R are used to do GIMME analysis

Thank you!

TANG Yingying 唐莹莹

Email: yingyingtang@utexas.edu

